

Supplementary Material

Supplementary Text: Common Abbreviations and Terms

AAs: amino acid sequences

3Di: Three-dimensional (3D) characters, derived from the Foldseek program's *structureto3didescrptor* function, applied to protein structure models in PDB or CIF format.

Note: The below terminology of Internode Certainty (IC), Tree Certainty, and related metrics is summarized from Kobert et al. (2016).

Bipartition. Imagine taking a phylogeny graphic, and erasing a branch *b*, changing nothing else. The erasing of *b* creates two subtrees, each with a set of taxa. Following Kobert et al. (2016), this set of two lists of taxa is notated as bipartition $B(b)$.

Relative frequency. For a branch *b* and bipartition $B(b)$ in a collection of phylogenies (e.g., bootstrap replicates or Bayesian posterior tree samples), the *relative frequency* is the ratio of the frequency of $B(b)$ in the tree collection to (the frequency of $B(b)$ + the frequency of B^*), where B^* is "the bipartition from the tree collection that has the highest frequency of occurrence and is incompatible with $B(b)$." (Kobert et al. 2016). This is notated as $X_{B(b)}$.

IC: Internode Certainty. The IC score is Shannon's measure of entropy, applied to the relative frequency of branch *b*. (See formula in Kobert et al. 2016.) This score ranges from 0 to 1. For example, if the relative frequency of *b* is 0.5, then, in R code,

$$IC = 1 + (0.5) * \log(0.5, \text{base}=2) + (0.5) * \log(0.5, \text{base}=2) = 0.0$$

If the relative frequency of *b* is 0.99, then in R code,

$$IC = 1 + (0.99) * \log(0.99, \text{base}=2) + (0.01) * \log(0.01, \text{base}=2) = 0.9192069$$

ICA: Internode Certainty All. This score is similar to IC, but attempts to include $B(b)$ plus *all* conflicting bipartitions that conflict with $B(b)$ and also with each other. This set is denoted as $C^*(b)$. However, finding this set is NP-hard. Therefore, an approximation is used, $C(b)$, which includes bipartitions with greater than 5% frequency. Once this set is defined, the ICA is just the Shannon information ($1 + (\text{frequency} * \log_2(\text{frequency})) \dots$) for all relative frequencies of all bipartitions in $C^*(b)$. This also ranges from 0-1.

TC: Tree Certainty. The IC, summed across all internal branches in the tree. (Terminal branches are trivial, as they are present in all trees with frequency 1.0, so they are excluded from the calculation.)

Relative TC. The TC, divided by the number of internal branches. I.e., the average IC across all internal branches.

TCA: Tree Certainty All. The ICA, summed across all internal branches in the tree.

Relative TCA. The TCA, divided by the number of internal branches. I.e., the average ICA across all internal branches.

Note: For IC and TC, if $B(b)$ is *not* the bipartition with the highest relative frequency in the tree collection, the score is multiplied by -1, to indicate that the tree collection positively supports a different topology.

References

Kobert, Kassian; Salichos, Leonidas; Rokas, Antonis; Stamatakis, Alexandros (2016). Computing the Internode Certainty and Related Measures from Partial Gene Trees. *Molecular Biology and Evolution*, 33(6), 1606-1617. <https://doi.org/10.1093/molbev/msw040>